

Student Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class / Set: \_\_\_\_\_

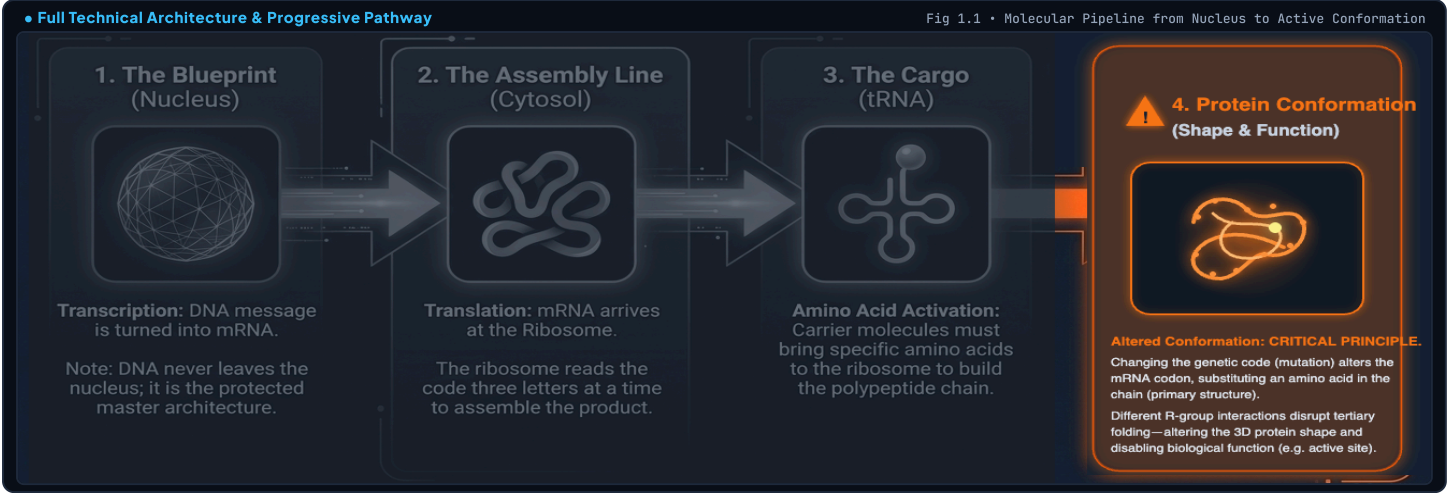
Target Grade: \_\_\_\_\_

RULE

THE CENTRAL DOGMA FRAMEWORK: STRUCTURE DETERMINES FUNCTION

Level 3 Essential Standard

To operate at a Level 3 standard, we must understand **molecular causality**: altering the DNA triplet code changes the amino acid sequence, directly disrupting tertiary folding, active site geometry, and biological function.



• ALTERED CONFORMATION: CRITICAL PRINCIPLE

Why Structure Strictly Dictates Function

Changing the genetic code (mutation) alters the **mRNA codon**, substituting a different amino acid into the polypeptide chain (**primary structure**). Different R-group chemical properties disrupt **tertiary folding** (hydrogen bonds, ionic bonds, disulfide bridges)—altering the 3D protein shape and disabling biological function (e.g. deforming the enzyme active site).

• The Level 3 Molecular Causality Chain (Memorise for Full Marks)

Sequential Step-by-Step Mechanism

STEP 01

DNA Mutation

Base substitution alters triplet code sequence in the gene.

STEP 02

Altered mRNA

Transcription produces mRNA with an altered codon.

STEP 03

Different Amino Acid

tRNA delivers a substituted amino acid during translation.

STEP 04

R-Group Disruption

New R-group cannot form original ionic / hydrogen / disulfide bonds.

STEP 05

Altered Tertiary Shape

3D conformation folds abnormally, changing active site geometry.

STEP 06

Loss of Function

Substrate can no longer bind; zero enzyme-substrate complexes form!

• Disulfide Bridges

Covalent bonds between sulfur atoms in two cysteine R-groups. Strongest tertiary bond; stabilizes conformation.

• Ionic Bonds

Electrostatic attractions between oppositely charged R-groups (-NH<sub>3</sub><sup>+</sup> and -COO<sup>-</sup>). Easily broken by pH changes.

• Hydrogen Bonds

Numerous weak attractions between polar R-groups (e.g. -OH and -NH). Individually weak, collectively vital; heat sensitive.

• Hydrophobic Forces

Non-polar R-groups fold inwards away from aqueous cytoplasm, driving spontaneous 3D compact globular folding.

• Level 3 Exam Application Task

6 Marks • High Yield Exam Question

Question: A patient carries a missense mutation involving a single base substitution in the gene encoding an essential metabolic enzyme. Explain step-by-step why this mutation causes the enzyme to lose its catalytic activity.

1. Effect on transcription:

(DNA triplet → mRNA codon) [1 Mark]

2. Primary structure & tRNA:

(Substituted amino acid in chain) [1 Mark]

3. R-group chemical bonding:

(Disrupted ionic, hydrogen, or disulfide bonds) [1 Mark]

4. 3D Tertiary folding:

(Abnormal folding & altered active site cleft) [1 Mark]

5. Substrate complementarity:

(Substrate no longer fits active site geometry) [1 Mark]

6. Biological & catalytic outcome:

(Zero enzyme-substrate complexes & loss of function) [1 Mark]

• Level 3 Examiner Mark Scheme Criteria (Tick Once Addressed in Your Answer)

Self-Assessment Checklist

☐ Stated DNA mutation changes the transcribed mRNA codon.

☐ Named disrupted bonding (hydrogen, ionic, or disulfide bridges).

☐ Explained active site is no longer complementary to substrate.

☐ Stated different amino acid alters primary structure sequence.

☐ Stated tertiary (3D) structure / conformation is altered.

☐ Concluded no enzyme-substrate complexes form → loss of activity.

Examiner's Top Tip for Distinction (Level 3): Never simply write "the protein doesn't work". Always link the chain: *Primary structure → R-group bonds → 3D tertiary conformation → active site complementarity → inability to form enzyme-substrate complexes.*